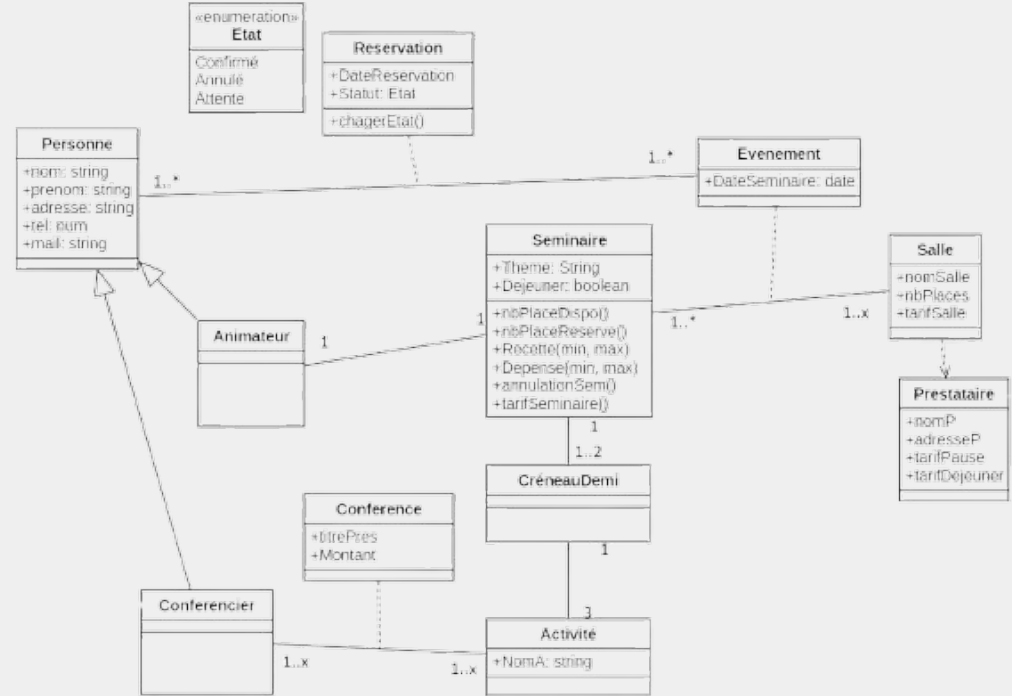


Association Relationship

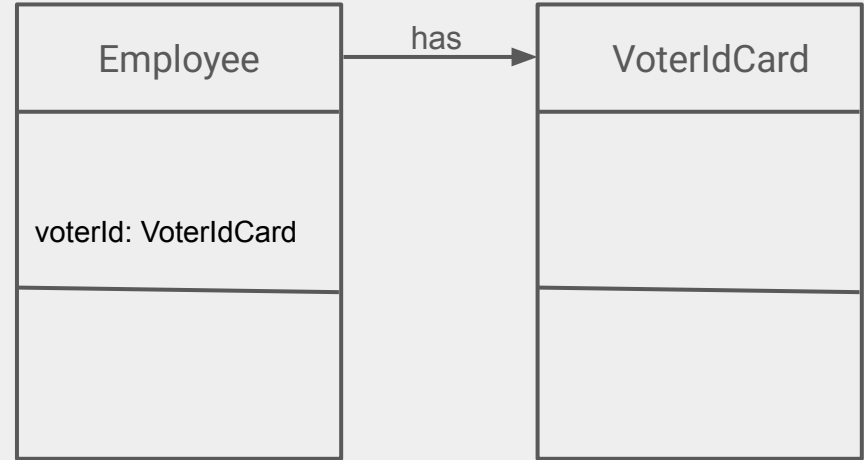
1. Composition
2. Aggregation
3. Delegation
4. Usage relationship



Association: One-One

The company requires a system to manage employee details, including their personal and government-issued identification records. Each employee must have a Voter ID Card to verify their age and eligibility for employment. Employees are eligible if they are at least 14 years old, as confirmed by their date of birth on the Voter ID.

The system must track each employee's total leave (20 days annually) and allow for the calculation of remaining leave based on the number of days taken. Additionally, the system should compute the employee's years of experience using their date of joining.



Association: One-One (Cont'd)

```
public class Employee {  
  
    public Employee ()  
    {  
        TotalLeave = 20;  
    }  
  
    public int TotalLeave {get; private set;}  
    public int RemainingLeave {get; private set;}  
    public string EmployeeId { get; set; }  
    public DateTime DateOfJoining { get; set; }  
    public string Designation {set;get;}  
    public string PresentAddress { get; set; }  
    public VoterIdCard VoterId { get; set; }  
  
    public void TakeLeave(DateTime from, DateTime to) {  
        RemainingLeave = TotalLeave - (to - from).Days;  
    }  
  
    public int GetExperience()  
    {  
        return DateTime.Now.Year - DateOfJoining.Year;  
    }  
  
    public int GetAge()  
    {  
        return DateTime.Now.Year - VoterId.DateOfBirth.Year;  
    }  
  
    public bool IsEligibleForEmployment() {  
        return (DateTime.Now.Year - VoterId.DateOfBirth.Year) >= 14;  
    }  
}
```

```
public class VoterIdCard {  
    public string Id { get; set; }  
    public string Name { get; set; }  
    public string FathersName {get;set;}  
    public string MothersName {set;get;}  
    public string VotingAddress {set;get;}  
    public DateTime Issuedate {set;get;}  
    public string BloodGroup {set;get;}  
    public DateTime DateOfBirth { get; set; }  
  
    public bool IsEligibleToVote()  
    {  
        return (DateTime.Now.Year - DateOfBirth.Year) >= 18;  
    }  
}
```

Association: One-One (Cont'd)

```
Employee employee1 = new Employee();
employee1.EmployeeId = "98083-0383";
employee1.DateOfJoining = new DateTime(2021, 1, 1);
VoterIdCard voterId = new VoterIdCard();
voterId.Id = "938-39378-393";
voterId.Name = "Dolon Khan";
voterId.DateOfBirth = new DateTime(1988, 1, 1);
employee1.VoterId = voterId;
employee1.TakeLeave(new DateTime(2021, 1, 1), new DateTime(2021, 1, 3));
Console.WriteLine(employee1.VoterId.Name);
Console.WriteLine(employee1.IsEligibleForEmployment());
Console.WriteLine(employee1.GetAge());
Console.WriteLine(employee1.GetExperience());
Console.WriteLine(employee1.VoterId.IsEligibleToVote());
Console.WriteLine(employee1.TotalLeave);
Console.WriteLine(employee1.RemainingLeave);
```

Practice 10: 1-1 Association

(Duration: 15 Min)

The bank needs a system to manage customer profiles along with their associated credit cards. Each customer should have a credit card that includes details such as the card number (must be digit), expiration date, and credit limit.

The system must verify if a credit card is valid by checking its expiration date. Additionally, the system should track a customer's available credit after purchases and allow for the calculation of outstanding balance based on the credit limit (500K) and spending.



Practice 11: 1-1 Association

(Duration: 15 Min)

The vehicle registration authority requires a system to manage car details and their associated license plates.

Each car must be assigned a unique license plate that includes attributes like the plate number, registration date, and expiration date.

The system must validate the license plate by ensuring it is current and not expired. Additionally, the system should track the car's owner information, manufacturer, model, and year of manufacture.

It must also calculate the age (twenty years or less) of the car to determine if it qualifies for registration renewal .



Association Relationship: 1- Many

The university requires a system to manage departments and the students admitted into them. Each department can have multiple students, with a defined maximum limit (200) on how many students can be admitted, but each student belongs to only one department.

The system should track department details such as the department name, head of the department, maximum student capacity, and the list of admitted students.

For each student, the system should store information like the student ID, name, date of admission.

The system must ensure that departments do not exceed their student capacity, allow for adding or removing students, and provide the ability to generate a list of students for each department.



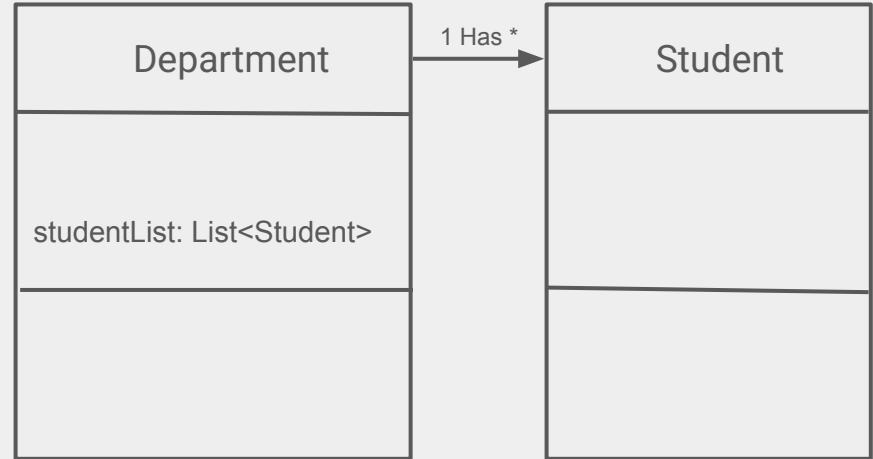
Association Relationship: 1- Many

```
public class Student{
    //properties
    //methods
}

public class Department{
    public List<Student> Students { get; set; }
    public Department()
    {
        Students = new List<Student>();
    }
    //properties
    //methods
}

//Client code

Student student1 = new Student();
Student student2 = new Student();
Department department = new Department();
department.Students.Add(student1);
department.Students.Add(student2);
Console.WriteLine(department.Students.Count);
```



Practice 12: 1- Many

(Duration: 30 Min)

A hospital requires a system to manage **doctors** and the **patients** under their care. **Each doctor can manage multiple patients, while each patient is assigned to only one primary doctor.** The system must allow doctors to perform various tasks, including **scheduling appointments, diagnosing medical conditions, and prescribing treatments to their patients.**

For instance, doctors can use the system to schedule appointments on specific dates for each patient and add new diagnoses to a patient's medical history. Additionally, doctors can assign treatments to patients based on their diagnosis and **discharge patients once they have recovered.**

On the patient's side, the system will allow patients to **view their upcoming appointments**, check if they have any ongoing treatments, and track their medical history, which is updated whenever the doctor provides a diagnosis. The system ensures that doctors can efficiently manage multiple patients while providing individualized care and keeping all medical records up-to-date.

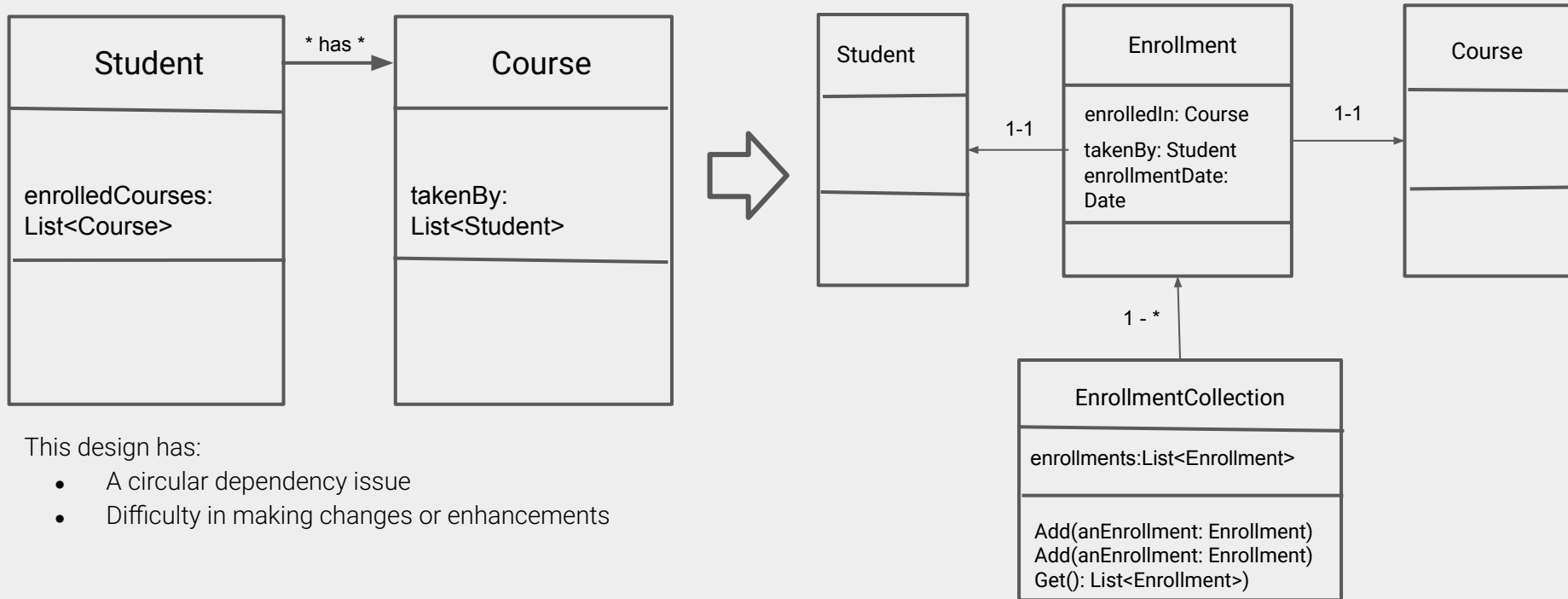


Association Relationship: Many - Many

In a large university, students have the opportunity to enroll in a wide variety of courses each semester. Each student can enroll in multiple courses, depending on their interests, degree requirements, and availability. At the same time, each course can be taken by many different students



Association Relationship: Many- Many



This design has:

- A circular dependency issue
- Difficulty in making changes or enhancements

Practice 13: Many - Many

(Duration: 30 Min)

In a hospital, patients are prescribed multiple medications, and each medication can be prescribed to many patients.

* Some examples of medications:

- ☐ Anti-Inflammatories
- ☐ Muscle Relaxants
- ☐ Antibiotics
- ☐ Cough Syrup
- ☐ Insulin
- ☐ Anticoagulants
- ☐ CNS Depressants
- ☐ Statins

Please ensure that following medications can't be prescribed combinedly:

1. Antibiotics & Statins
2. Muscle Relaxants & CNS Depressants
3. Anti-Inflammatories & Anticoagulants

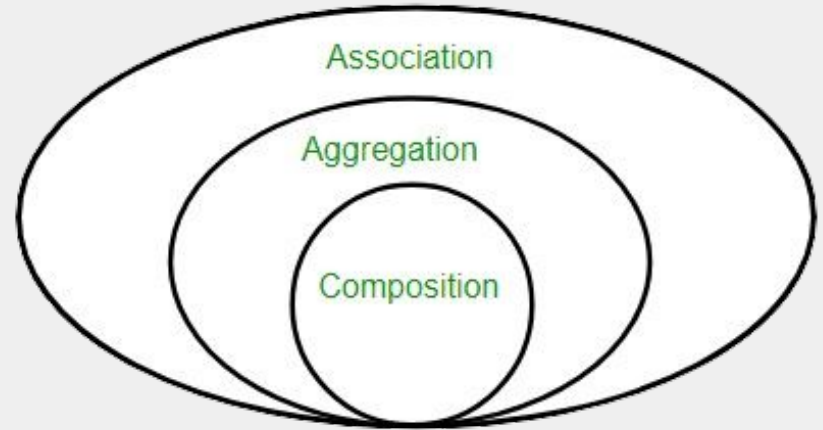


Composition, Aggregation

An association is a general term that describes the relationship between two independent classes.

Aggregation is a more specific type of association where one class, known as the whole, contains a collection of other classes. For example, a library and its books demonstrate aggregation, as the books can exist independently of the library.

Composition is a stronger form of aggregation that implies ownership and lifecycle dependency. If the whole is destroyed, the parts cease to exist. A common example is a house and its rooms; a room cannot exist without the house.



Delegation

“Delegation” refers to a design where an object hands off or delegates a specific task or responsibility to another object. It's actually an association relationship. Benefits of Delegation are:

- **Flexibility:** Can be changed easily, if needed.
- **Code Reuse:** Different classes can delegate common tasks to a shared delegate, reducing duplication.
- **Separation of Concerns:** The delegating object is simplified and focuses on high-level tasks, while the delegate handles specific behaviors.



Usage Relationship

"Usage" relationship refers to a situation where one class temporarily uses another class to perform some operation or achieve a specific task.

- The relationship is temporary or on-demand.
- It usually occurs through method parameters, local variables, or temporary objects created within a method.

```
class ShoppingCart:
    def checkout(self, amount):
        payment_processor = PaymentProcessor()
        success = payment_processor.process_payment(amount)

        if success:
            print("Payment processed successfully.")
        else:
            print("Payment failed. Please try again.")

class PaymentProcessor:
    def process_payment(self, amount):
        print(f"Processing payment of ${amount}...")
        return True

if __name__ == "__main__":
    cart = ShoppingCart()
    cart.checkout(100.50)
```


More practices of Association Relationship

Smart Parking System

In a smart parking lot, the system must manage vehicle entry, slot assignment, payment, and exit.

- Each vehicle will be assigned a specific parking slot.
- When a vehicle leaves, the parking charge will be calculated based on the duration of stay.
- Slot allocation and availability must be updated in real time.

Restaurant Order Management System

In a restaurant, customers place orders where each order can include multiple menu items, and the kitchen prepares those orders.

- The total bill must be calculated for each order.
- The kitchen can update the order status (Pending, Cooking, Served).
- Once the customer confirms the order, the payment will be completed.

Library Management

A library contains many books, and each book can have one or more copies.

- Members can borrow books for a specific period.
- Regular members can borrow up to 3 books, while Premium members can borrow up to 5.
- If a book is returned late, a fine of 10 taka per day will be added.
- For Premium members, the fine rate will be half.